

ECB Instructions for Reporting the Validation Results of Internal Models

Methodological Reflections and Opportunities for Further Enhancement

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ECB Supervisory Monitoring

- To facilitate supervisory monitoring through a comparable quantitative framework, the ECB issued the [Instructions for Reporting the Validation Results of Internal Models – IRB Pillar I Models for Credit Risk](#).
- The document explains the rationale for the additional supervisory reporting and introduces a common set of validation tools, statistical tests, and analyses that enable the ECB to perform a more consistent initial assessment of model performance across institutions.
- The instructions explicitly acknowledge that they are not legally binding and do not replace or override applicable European or national legislation.
- Furthermore, the ECB emphasizes that the prescribed validation tools should not be regarded as the minimum set of analyses required to demonstrate regulatory compliance. Instead, institutions remain responsible for designing validation frameworks that are proportionate to the complexity, materiality, and intended use of their internal models.
- Nevertheless, institutions supervised by the ECB are expected to report the results of the prescribed validation tools as part of the additional supervisory information requested under individual supervisory decisions.

ECB Supervisory Monitoring cont.

- Regarding the interpretation of the submitted results, the ECB also recognizes that statistical validation techniques have inherent limitations and rely on assumptions that may not always hold in practice, particularly for small samples. However, no other explicit limitations are discussed.
- Given that banks typically regard ECB documents as one of the most influential references when designing their modeling frameworks, it is worth asking whether certain approaches presented in this document should be further challenged and their limitations more thoroughly explored.
- The purpose of this presentation is to provide a methodological reflection on specific validation tests and potentially open up a discussion on opportunities for further enhancements. Specifically, the following validation approaches are highlighted in this presentation:
 - 1 Concentration in rating grades in Probability of Default (PD) models;
 - 2 Predictive ability testing in Probability of Default (PD) models;
 - 3 Predictive ability testing in Loss Given Default (LGD) and Exposure at Default (EAD) models;
 - 4 Somers' D as a measure of discriminatory power in Loss Given Default (LGD) and Exposure at Default (EAD) models.

Concentration in Rating Grades in PD Model

ECB Proposed Approach

In the [Instructions for reporting the validation results of internal models](#), the ECB requires banks to report the Herfindahl Index (HI) value for the current application portfolio, along with the p-value of statistical tests used to assess changes in the Herfindahl Index from the model's development phase to its current application.

The formula for the Herfindahl Index is provided as follows:

$$HI = 1 + \frac{\log\left(\frac{CV^2 + 1}{K}\right)}{\log(K)}$$

with CV being a coefficient of variation:

$$CV = \sqrt{K \sum_{i=1}^K \left(R_i - \frac{1}{K}\right)^2}$$

where:

- K is the number of rating grades;
- R_i is the relative frequency of the i -th rating grade.

Concentration in Rating Grades in PD Model cont.

ECB Proposed Approach cont.

The change in the HI is assessed indirectly by testing the change in the CV using a normal approximation and the assumption of a deterministic CV from the model's development phase.

The test's null hypothesis is that the HI at the time of the model application is lower than or equal to the HI at the time of development. Based on these assumptions, the p-value is calculated as follows:

$$1 - \Phi \left(\frac{\sqrt{K-1}(CV_{curr} - CV_{init})}{\sqrt{CV_{curr}^2 (0.5 + CV_{curr}^2)}} \right)$$

where:

- Φ is the cumulative distribution function of the standard normal distribution;
- CV_{curr} is the coefficient of variation at the time of the model application;
- CV_{init} is the coefficient of variation at the time of model development (initial validation);
- K is the number of rating grades or bins of the risk factors.

Concentration in Rating Grades in PD Model cont.

Challenges Associated with the Proposed Approach

While the value of the HI is typically compared directly to specific thresholds - such as 0.20, 0.25, and 0.30 - the change in HI is assessed indirectly through variations in the CV. This raises two key questions: First, does a change in the relative dispersion measure reflect changes in the HI value? Second, how does the assumption of the deterministic CV_{init} influence the test results?

Consider the following data: relative frequencies at the time of the model's development are [0.13, 0.14, 0.23, 0.20, 0.11, 0.10, 0.09], and at the time of the model's application, they are [1, 0, 0, 0, 0, 0, 0]. This simulated data assumes that the HI shifts to a fully concentrated rating scale at the time of the model's application.

Let's examine the HI values during these two periods. The HI value at the time of the model's development is 0.057 while the HI value at the time of the model's application is 1.

Given the assumption of strong concentration, practitioners would expect the p-value for the test of change in the HI to be well below the commonly used significance levels of 0.01 or 0.05. However, the calculated p-value based on the proposed approach is 0.2043.

As observed, the calculated p-value is significantly higher than the commonly used significance levels, leading us to fail to reject the null hypothesis.

Further details on concentration testing across rating grades can be found [here](#).

Predictive Ability Testing in PD Models

ECB Proposed Approach

In its instructions, the ECB proposes the Jeffreys test for testing predictive ability. This is a commonly used test in practice and is typically applied to assess evidence of underestimation of the calibrated PD.

The p-value of the Jeffreys test is calculated as follows:

$$Pr(X \leq d) = \beta(q = PD, \text{shape1} = nd + 0.5, \text{shape2} = no - nd + 0.5)$$

where:

- β is the cumulative distribution function of the beta distribution;
- PD is the calibrated PD;
- nd is the number of defaults;
- no is the number of observations.

Finally, the ECB requires this test to be performed at both the portfolio level and the level of each rating grade.

Predictive Ability Testing in PD Models cont.

Challenges Associated with the Proposed Approach

The proposed approach raises two key questions:

- 1 the statistical validity of the proposed test when applied to portfolio-level data;
- 2 the alignment of the Jeffreys test with the nature of IRB models and its comparison with other statistical tests used for the same purpose.

When moving from rating-grade-level to portfolio-level testing, it is important to recognize that portfolio (segment) data consist of heterogeneous obligors assigned PD estimates derived from homogeneous rating grades. Moreover, heterogeneity in PD estimates across rating grades is a regulatory requirement for IRB models.

Neglecting this feature and applying the Jeffreys test as proposed may affect the statistical validity of the test. Specifically, it affects the variance calculation, as the variance may be overestimated when the underlying structure of the PD model is not taken into account. The same issue applies to the other two statistical tests commonly used for PD predictive ability testing: the z-score test and the exact binomial test. However, for these two frequentist tests, there are direct ways to adjust for heterogeneity in PD estimates without changing the underlying assumptions or testing framework. In contrast, because the Jeffreys test is based on a Bayesian approach, a direct adjustment for heterogeneity would alter the underlying assumptions of the test.

Further details on this topic can be found [here](#).

Predictive Ability Testing in PD Models cont.

Challenges Associated with the Proposed Approach cont.

As already mentioned, the Jeffreys test belongs to the Bayesian framework and is used for interval estimation of a binomial proportion, whereas the other tests are based on the frequentist framework. This raises the question of which approach is more suitable given the nature of PD IRB models.

When hard thresholds are used to assess model predictive ability, the Jeffreys test and other tests applied for the same purpose may lead to different conclusions. In such cases, it is important to consider how these differences should be interpreted and addressed. At a minimum, practitioners should be informed about the underlying assumptions of each test, the hypotheses being tested, and the potential limitations of the proposed approach.

Further details on the differences between statistical approaches to PD predictive ability testing can be found [here](#).

Predictive Ability Testing in LGD and EAD Models

LGD Models - ECB Proposed Approach

In its instructions, the ECB defines the t-test as a one-sample test for paired observations, which compares estimated LGD with realised LGD under the null hypothesis that the estimated LGD is greater than the true LGD (one-sided hypothesis test), assuming independent observations.

The test statistic is given as follows:

$$T = \sqrt{N} \frac{\frac{1}{N} \sum_{i=1}^N (LGD_i^R - LGD_i^E)}{\sqrt{s_{LGD}^2}}$$

$$s_{LGD}^2 = \frac{\sum_{i=1}^N \left((LGD_i^R - LGD_i^E) - \frac{1}{N} \sum_{j=1}^N (LGD_j^R - LGD_j^E) \right)^2}{N - 1}$$

where:

- N is the number of facilities used for testing;
- LGD_i^E denotes the estimated LGD for facility i ;
- LGD_i^R denotes the realised LGD for facility i .

Predictive Ability Testing in LGD and EAD Models cont.

LGD Models - Challenges Associated with the Proposed Approach

Given the proposed approach and the fact that, for discrete rating-scale facility grades or pool estimates, the estimates are single values (e.g., long-run averages or final estimates), an important question arises: what exactly is this procedure testing?

When applied to a discrete rating scale at the facility-class level, this approach effectively assumes that the true LGD is represented by the value observed in the validation sample. Therefore, several straightforward questions arise:

- How does this assumption align with the standard statistical hypothesis-testing framework?
- How does this assumption align with the principles applied in other backtesting exercises?
- How does this assumption align with the nature of IRB models?

The choice of statistical assumptions directly affects the interpretation of predictive ability results. Therefore, at a minimum, practitioners should be informed about the implications of attributing the entire uncertainty to the validation sample while treating the calibrated estimate as fixed. Moreover, the validation framework should be aligned as closely as possible with the Through-the-Cycle nature and inherent conservatism of IRB LGD estimates.

Further details on this topic can be found [here](#).

Predictive Ability Testing in LGD and EAD Models cont.

EAD Models - ECB Proposed Approach

In its instructions, the ECB defines the t-test as a one-sample test for paired observations, which compares estimated CCF with realised CCF under the null hypothesis that the estimated CCF is greater than the true CCF (one-sided hypothesis test), assuming independent observations.

The test statistic is given as follows:

$$T = \sqrt{R} \frac{\sum_{i=1}^R (CCF_i^R - CCF_i^E)}{\sqrt{s_{CCF}^2}}$$

$$s_{CCF}^2 = \frac{\sum_{i=1}^R \left((CCF_i^R - CCF_i^E) - \frac{1}{R} \sum_{j=1}^R (CCF_j^R - CCF_j^E) \right)^2}{R - 1}$$

where:

- R is the number of facilities used for testing;
- CCF_i^E denotes the estimated LGD for facility i ;
- CCF_i^R denotes the realised LGD for facility i .

Predictive Ability Testing in LGD and EAD Models cont.

EAD Models – Challenges Associated with the Proposed Approach

The proposed approach should essentially be consistent with the approach used for LGD backtesting. However, the formula for CCF backtesting using the t-test does not reflect this. Specifically, there is an error in the numerator of the test statistic: instead of the sum of the differences between the realised and estimated values, the mean of these differences should be used.

In addition to the aforementioned formula error, the challenges applicable to LGD backtesting are equally relevant to the proposed CCF backtesting approach. This is particularly evident for CCF models, where discrete facility-level scales are commonly used by practitioners.

Discriminatory Power Testing in LGD and EAD Models

ECB Proposed Approach

In its instructions, the ECB defines the generalized AUC (gAUC) as a required measure of discriminatory power for LGD and EAD models. Furthermore, gAUC is defined in terms of Somers' D, while the details of the Somers' D calculation are provided in the section titled "3.2. Annex 2: Statistics on the Discriminatory Power of LGD/CCF Models".

The presented formula defines Somers' D(C|R), which assumes that the row variable, LGD^E , is the independent variable, while the column variable, LGD^R , is the dependent variable.

Discriminatory Power Testing in LGD and EAD Models

cont.

Challenges Associated with the Proposed Approach

In his paper, Robert H. Somers (1962) proposed a pair of asymmetric coefficients for measuring association in ordered contingency tables. Essentially, Somers distinguishes between situations in which one variable is considered independent and the other dependent. For a contingency table (X, Y) , the coefficient D_{yx} reflects the situation where X is the independent variable and Y is the dependent variable.

The approach required by the ECB calculates the D_{yx} version of Somers' D. However, this version is not typically considered a measure of the discriminatory power of X as a predictor of Y .

A direct interpretation of the pair of Somers' D coefficients is provided in Newson, R. (2006), [Confidence Intervals for Rank Statistics: Somers' D and Extensions](#). Newson interprets the pair of Somers' D coefficients as follows:

- 1 D_{yx} represents an effect size, measuring the effect of X on Y ;
- 2 D_{xy} represents a predictor performance indicator, measuring the performance of X as a predictor of Y .

Aligning the common use of Somers' D for calculating gAUC with regulatory reporting requirements is of utmost importance, as differences between the estimated metrics can be substantial and may lead to significantly different conclusions about the quality of risk differentiation functions in these models.

Further details on this topic can be found [here](#).

Conclusions

- The ECB Instructions explicitly state that the prescribed validation tools should not be regarded as the minimum regulatory requirements, while institutions remain responsible for designing validation frameworks proportionate to the complexity, materiality, and intended use of their internal models.
- Nevertheless, the Instructions represent a regulatory reference providing an overview of concrete statistical approaches for IRB model validation and significantly influence banks' validation practices.
- The methodological reflections presented highlight potential challenges related to the statistical validity, underlying assumptions, and interpretation of selected validation approaches, including concentration testing, predictive ability testing, and discriminatory power assessment.
- Practitioners should be adequately informed about the assumptions and limitations of the prescribed tests and assess their alignment with the specific characteristics of IRB models.
- These observations provide opportunities for further methodological discussion and enhancement of existing validation practices.